

I. Link Definition and Desired Margin

- a. **Next communication node:** the nearer of the two orbital relay agents. The rover does not run its mission through a direct-to-Earth link; Section V shows why. Uplink is taken in the assignment's sense of rover $\rightarrow$ relay (the science return direction), and downlink as relay $\rightarrow$ rover (the command forward direction).
- b. **Band and channel:** UHF proximity link on the CCSDS Proximity-1 return/forward pair, 401.6 MHz return and 437.1 MHz forward, the same split flown by Electra and Electra-Lite on the Mars Relay Network.
- c. **Desired margin:** 3 dB above the E_b/N_0 required for a post-decoding bit error rate of 10^{-6} . With CCSDS turbo coding at rate $1/2$ that threshold is ≈ 1.5 dB, so the link must deliver at least 4.5 dB. Three decibels is the right allowance here because the relay orbit geometry, the helix pattern at the rover, and the Jovian synchrotron contribution to receiver noise are all only coarsely known before flight.
- d. **Worst-case geometry:** both relays fly a 200 km circular Europa orbit. With $R_{\text{Europa}} = 1561$ km and a 10° elevation mask, the slant range at the edge of the pass is

$$R = -R_E \sin \varepsilon + \sqrt{R_E^2 \sin^2 \varepsilon + (r^2 - R_E^2)} = 588 \text{ km},$$

against 200 km at zenith. Every budget below uses the 588 km worst case.

II. Assumed Parameters

- a. $P_t = 5$ W RF (rover), 8 W RF (relay). The rover's UHF radio draws 35 W DC in the power budget; an Electra-Lite class transceiver puts roughly 5 W of that on the antenna. The relay agent is not power-constrained the way the rover is, so it carries a larger amplifier.
- b. $G_t = 3.16$ (5.0 dBi) rover quadrifilar helix; $G_r = 5.01$ (7.0 dBi) relay UHF patch array. The helix is the antenna already carried in the mass budget, and its near-hemispherical pattern is what lets the rover close a pass without pointing.
- c. $L_a = 0.89$ (−0.51 dB). Europa has no meaningful atmosphere at $\sim 10^{-12}$ bar, so this term is not atmospheric absorption. It covers circular-to-linear polarization mismatch and Jovian plasma scintillation on the path.
- d. $L_l = 0.71$ (−1.49 dB) transmitter-to-antenna line loss, from cabling and connectors run to a mast-mounted antenna and derated for cryogenic operation at 90 K.
- e. $\lambda = 0.7465$ m return (401.6 MHz), $\lambda = 0.6859$ m forward (437.1 MHz), from $\lambda = c/f$.
- f. $k = 1.38 \times 10^{-23}$ J/K.
- g. $T_s = 3000$ K at the relay receiver, 4000 K at the rover receiver. This is the parameter that separates a Europa proximity link from a Mars one. At 400 MHz the receiver sees galactic background plus Jupiter's synchrotron emission, and the rover's front end has no cryogenic cooling and a warmer view of the source, so it is assigned the higher figure.
- h. $R = 5.879 \times 10^5$ m, the 10° -elevation slant range derived above.

- i. $R_b = 128$ kbps return, 8 kbps forward. These are the floors set by REQ.10.01: 55 MB of compressed data per sol delivered inside the 90 min relay window the ConOps allocates works out to 90 kbps, and 128 kbps is the next standard step up.

III. Method

- a. Ratio form:

$$\frac{E_b}{N_0} = \frac{P_t G_t G_r L_a L_l \lambda^2}{(4\pi R)^2 k T_s R_b}$$

- b. Decibel form: $(E_b/N_0)_{\text{dB}} = 10 \log_{10} (E_b/N_0)$

- c. Margin: $M = (E_b/N_0)_{\text{dB}} - (E_b/N_0)_{\text{req}}$, with $(E_b/N_0)_{\text{req}} = 1.5$ dB.

- d. The link closes when $M \geq 3$ dB.

IV. Uplink — Rover to Relay (Science Return)

- a. Numerator:

$$P_t G_t G_r L_a L_l \lambda^2 = 5 \times 3.16 \times 5.01 \times 0.89 \times 0.71 \times (0.7465)^2 = 27.87$$

- b. Denominator:

$$(4\pi R)^2 k T_s R_b = \left(4\pi \times 5.879 \times 10^5\right)^2 \times 1.38 \times 10^{-23} \times 3000 \times 1.28 \times 10^5 = 0.2892$$

- c. Result: $E_b/N_0 = 27.87/0.2892 = 96.4$

- d. In decibels: $10 \log_{10}(96.4) = 19.84$ dB

- e. Margin: $19.84 - 1.5 = 18.3$ dB. **The uplink closes.**

Term-by-term, in decibels, for traceability:

Term	Value (dB)
P_t (5 W)	+6.99
G_t (rover helix)	+5.00
G_r (relay patch array)	+7.00
L_a (polarization + plasma)	−0.51
L_l (line loss)	−1.49
λ^2	−2.54
$(4\pi R)^2$ (space loss at 588 km)	−137.37
k	+228.60
T_s (3000 K)	−34.77
R_b (128 kbps)	−51.07
E_b/N_0	+19.84

V. Downlink — Relay to Rover (Command Forward)

a. Numerator:

$$8 \times 5.01 \times 3.16 \times 0.89 \times 0.71 \times (0.6859)^2 = 37.65$$

b. Denominator:

$$\left(4\pi \times 5.879 \times 10^5\right)^2 \times 1.38 \times 10^{-23} \times 4000 \times 8.0 \times 10^3 = 0.02410$$

c. Result: $E_b/N_0 = 37.65/0.02410 = 1562$

d. In decibels: $10 \log_{10}(1562) = 31.94 \text{ dB}$

e. Margin: $31.94 - 1.5 = 30.4 \text{ dB}$. **The downlink closes.**

The forward direction has an easier job: the relay transmits more power from a higher-gain antenna, and the command load is 8 kbps against 128 kbps of science return. The $16 \times$ rate difference alone accounts for 12 dB of the gap.

VI. Direct-to-Earth Check (Why the Relay Exists)

The same equation run on the rover's X-band low-gain antenna against a DSN 34 m BWG station sets the fallback rate. Using $P_t = 15 \text{ W}$, $G_t = 6.31$ (8 dBi LGA), $G_r = 6.6 \times 10^6$ (68.2 dBi), $\lambda = 0.0357 \text{ m}$ (8.4 GHz), $T_s = 30 \text{ K}$ (cryogenic DSN front end), and solving for the rate that still holds 4.5 dB:

Earth range	R (m)	Max rate at 4.5 dB
Opposition (4.2 AU)	6.28×10^{11}	6.9 bps
Conjunction (6.2 AU)	9.27×10^{11}	3.2 bps

Seven bits per second is a state-of-health beacon, not a science link: one 55 MB sol of REQ.10.01 would take just over two years to clear. This is the quantitative reason the relay agents are architectural rather than optional, and it is the number REQ.10.03 fixes as the 5 bps fallback floor.

VII. Reflection

Both directions close, and by a wide margin: 18.3 dB on the return and 30.4 dB on the forward. The interesting result is that the proximity link is not power-limited at all. At 4.5 dB the return would support roughly 4.4 Mbps, so the real ceiling is the radio's 2048 kbps maximum and the 90 min pass, not the RF budget. That surplus is worth keeping rather than spending, since T_s is the weakest assumption I made and Jupiter's synchrotron noise at 400 MHz could plausibly be several times my 3000 K. If the link did fail, the first parameter I would change is R_b , because it is free, and the second is G_r at the relay, since adding patch elements to an orbiter is cheaper in mass and power than anything on the rover.

References:

- [1] Jet Propulsion Laboratory, "DSN Telecommunications Link Design Handbook," 810-005, Rev. E, NASA/JPL, Pasadena, CA. [Online]. Available: <https://deepspace.jpl.nasa.gov/dsndocs/810-005/>.

- [2] Consultative Committee for Space Data Systems, “Proximity-1 Space Link Protocol — Physical Layer,” CCSDS 211.1-B-4, 2013.
- [3] Consultative Committee for Space Data Systems, “TM Synchronization and Channel Coding,” CCSDS 131.0-B-4, 2017.
- [4] C. D. Edwards *et al.*, “The Electra Proximity Link Payload for Mars Relay Telecommunications and Navigation,” Jet Propulsion Laboratory, Pasadena, CA.
- [5] NASA, “Deep Space Network,” NASA Space Communications and Navigation. [Online]. Available: <https://www.nasa.gov/communicating-with-missions/dsn/>.
- [6] NASA, “Radar for Europa Assessment and Sounding: Ocean to Near-surface (REASON),” Europa Clipper Mission. [Online]. Available: <https://europa.nasa.gov/spacecraft/instruments/reason/>.

AI disclosure: AI was used to verify the arithmetic in each link budget term and to check the assumed parameter values against published Electra and DSN performance figures. The architecture choice, parameter selection, and margin definition are mine.